

Beyond Answers: Rethinking AI Assistants to Support Students' Computational Thinking

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Hypothesis

Students now days rely heavily on AI assistants which undermines their ability to develop Computational Thinking (CT) skill that is required for independent *problem solving*.

Computational Thinking

Problem-solving approach that involves *breaking down* complex problems into manageable parts, *recognizing patterns*, designing step-by-step solutions (*algorithms*), and using *abstraction* to focus on relevant details.

Research Objectives

- ✓ Understand students study habits and patterns for using AI assistants.
- ✓ Identify key problems and challenges with the current use of AI in CT Education.
- ✓ Outline a set of AI tools (a framework) for mitigating those challenges.
- Build and test these tools.

Research Design

Surveys

100+ ICL students (under and postgraduate) on:

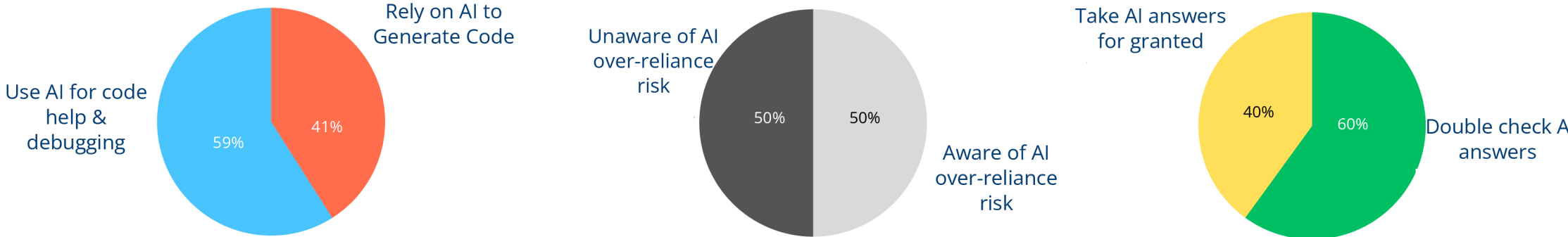
- Learning Behaviours
- AI Advantages and Risks
- Problem Solving Methods

Interviews

9 ESE postgraduate students on:

- Research Methods
- Awareness of AI Risks
- Peer Engagement

Results

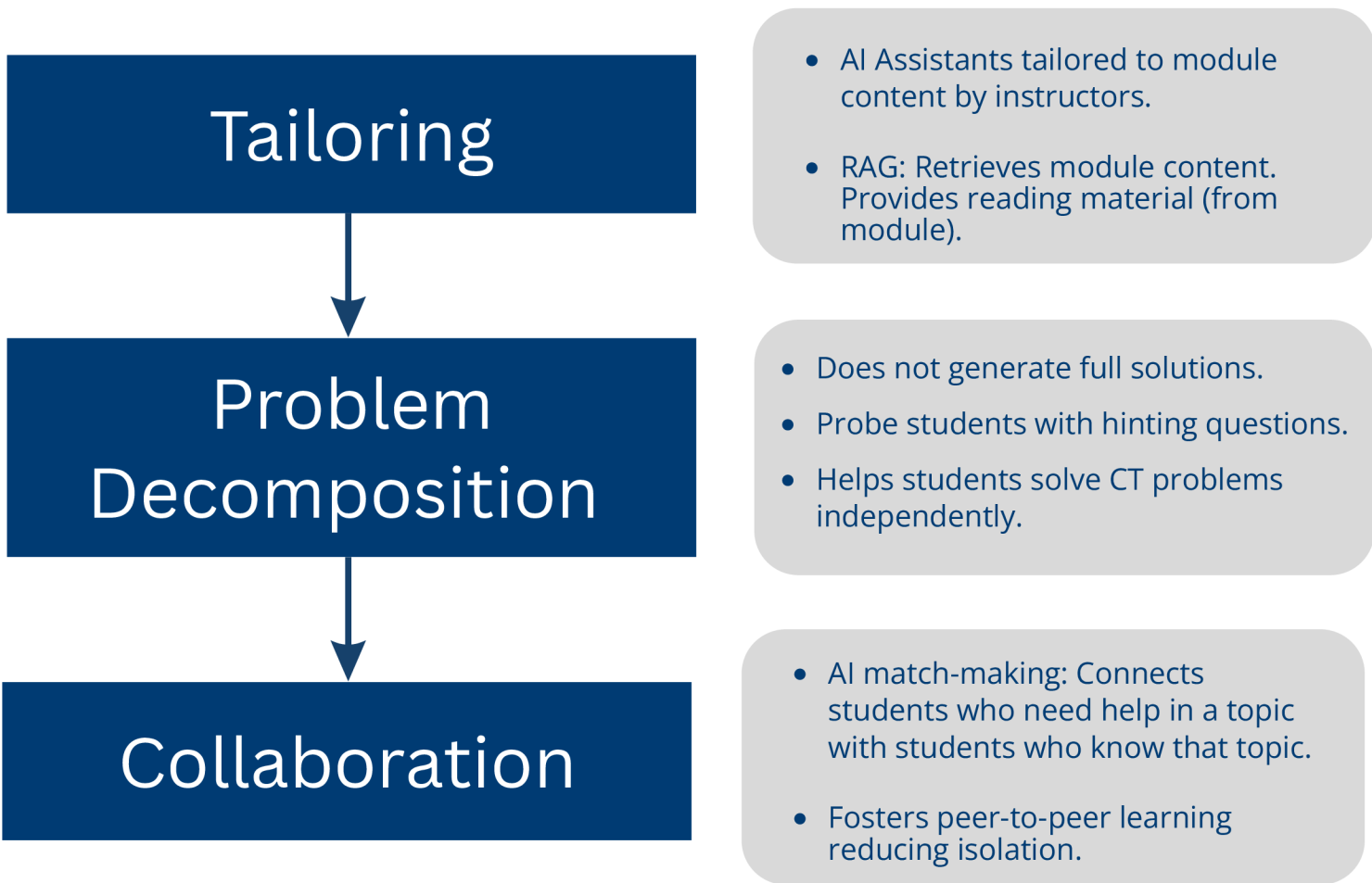


Identified Challenges

1. *Over-reliance*: Students develop over-reliance on AI assistants Undermining their CT skill development.
2. *Awareness*: Many students are unaware of this risk.
3. *Isolation*: Reduced instructor and peer interaction.
4. *Broad Assistance*: Students receive out-of-context or more-than-required information from generic AI assistants.

AI Framework

We proposed building a set of context-aware and CT supportive AI assistants:



CT AI Assistant

Hey, Alice!

Hey, whatsup?

I can see from the module board, you started taking search algorithms, do you need any help?

Yes, can you explain the binary search?

Sure, how are you with nested loops?

Not much..

Ok, [here](#) is explanation from your module slides on nested loops and [here](#) is the text book, give it a read and let me know if you're still struggling.

Ok, I grasped it

Perfect, lets move on, given this list of numbers, whats the number of steps needed to find "4"?

3?

[16, 42, 4, 15, 8]

Yes, anything we can do to the list to find "4" quicker?

mm sort it?

Exactly, [4, 8, 15, 16, 42] cutting down the steps to "1" Now that it is sorted, how many steps to find "16"?

mm 4?

Not necessarily, think outside the box.

...

Yo, AI

Yes Bob!

I am struggling to grasp binary search algorithm!

Alice connected!

A: Hey Bob, heard you need help understanding Binary Search?

B: Yes please!

A: Do you want to jump on a quick video call?

B: Sure, I'll call you